Deep Sweep Page — Information Document

Page Overview

Purpose

The Deep Sweep page runs a **one-off broad query across multiple repositories in parallel**. It is the multi-repository counterpart to the Deep Dive page: users pick a set of repositories, enter keywords and (optionally) a date range, cap the per-repository result count, and optionally enable AI summarization. The backend creates a `deep_sweep` execution, queries each repository in sequence, normalises and de-duplicates the combined result set (by DOI, and by title + first-author), generates a single Markdown report, and — if AI is enabled — produces a per-document summary that is embedded in the report.

Primary User Flows

1. **Configure and run.** Select repositories → enter keywords → optionally set date range, max-results cap, AI toggle, and per-execution AI overrides → click **Run Deep Sweep**.
2. **Load a saved configuration.** Use the `ConfigLoader` at the top of the form to reapply a previously saved `manual_sweep` configuration; the date range is intentionally excluded from saved configurations and must be entered fresh each run.
3. **Save the current form as a configuration.** Open the **Save Configuration** modal, give it a name, and save to the `manual_sweep` configuration set.
4. **Track and follow-up.** On submit, the page receives an `execution_id` and registers it with the shared `ExecutionContext`; live progress appears in the floating widget and on the Monitor page. When the run finishes, a result card with counts and a link to the report is shown.

Inputs and Outputs

Inputs (form state):

- `repositories` — list of repository slugs to query (required, ≥ 1).

- `keywords` — list of keyword tokens (required, ≥ 1); joined with spaces to form the underlying `query` string.
- `dateFrom` / `dateTo` — optional ISO date strings; sent as `null` when empty.
- `maxResults` — integer 10–500 (step 10, default 100), **per repository**.
- `aiEnabled` — boolean toggle.
- `aiOverrideProvider` / `aiOverrideModel` / `aiOverrideLength` / `aiOverrideTone` / `aiOverrideTemperature` / `aiOverrideExtractionGoals` (IMPL-AI13, full-parity) — optional per-execution overrides surfaced through the shared `AIOverridePanel` ; only non-empty values are included in
  the request body (per-field merge — empty values never clobber persisted defaults). The `Model` control is a provider-aware dropdown populated by `Load models` ; if the chosen provider has no stored API key the panel renders an inline key input that POSTs to `/api/credentials/{name}` , and a `Save as default model` button persists the selection into `ai_default_models[provider]` without leaving the page.
- `ephemeralKeys` — in-memory map of `{credential_name: key_value}` for keyed repositories whose key is not stored in the keyring.

Outputs:

- `POST /api/search/sweep` returns `{ execution_id: number }` .
- A `deep_sweep` execution row is persisted by `SweepEngine.prepare_execution` (with `parameters` JSON, `start_time` , and `execution_type = "deep_sweep"`).
- On completion: a Markdown report under `resmon_reports/markdowns/` and a per-task log under `resmon_reports/logs/` .
- Live progress events on the execution's progress channel (`execution_start` , `repo_start` , `query_progress` , `repo_done` , `dedup_stats` , `report_saved` , `complete` , etc.).

Known Constraints or Permissions

- **Admission controller.** Before launching, the backend calls `_reject_if_at_manual_cap()` ; if the concurrent-execution limit is reached, the request is rejected with **HTTP 429** and a `Retry-After: 5` header. The error message references **Settings** → **Advanced** where the cap is configurable.
- **Required keys.** Key-less repositories (e.g. arXiv, CrossRef, OpenAlex, bioRxiv, medRxiv) always work; the four repositories listed in `_REQUIRED_CREDENTIALS` (`core` , `ieee` , `nasa_ads` , `springer`) require a stored or ephemeral key. A missing key does not fail the sweep — the engine emits a `repo_skipped_missing_key` event, the task log records a warning, and that repository's query returns zero results.

- **At least one repository and one keyword** are required; client-side validation prevents submission otherwise.
- **Per-repository cap, not global.** `max_results` applies to each repository individually; the merged, de-duplicated result set may be smaller than `repositories.length × max_results`.
- **Save-as-configuration omits the date range** by design; the date range must be chosen fresh on every run.

Frontend

Route and Main Component

- Route: `/sweep` (mounted in `App.tsx`).
- Main component: `DeepSweepPage` in `resmon_scripts/frontend/src/pages/DeepSweepPage.tsx`.

Child Components and Hooks

Child components rendered inside the form card:

- `PageHelp` — collapsible on-page help with "When to use this page", "How to use it", and "Tips" sections.
- `ConfigLoader` (`configType="manual_sweep"`) — dropdown to load saved configurations; a `refreshKey` prop triggers a reload after a save.
- `RepositorySelector` (`mode="multi"`) — multi-select over the repository catalog.
- `KeywordCombinationBanner` — mounted under the selector when at least one repository is chosen; lists the upstream keyword-combination semantics for each selected repository (e.g. "Implicit AND", "Relevance-ranked"), with a tooltip pointing to the consolidated glossary on the Repositories & API Keys page.
- `RepoKeyStatus` (`variant="sweep"`) — rendered once per selected repository; shows whether the credential is stored, absent, or accepted as an ephemeral value, and offers inline entry for ephemeral keys.
- `DateRangePicker` — two date fields wired to `dateFrom` / `dateTo`.
- `KeywordInput` — chip-style keyword entry (Enter-to-commit).
- `InfoTooltip` — small contextual hints next to the max-results and AI toggles.

Hooks:

- `useExecution()` (from `ExecutionContext`) — provides `activeExecutions` and `startExecution(id, type, repositories)`.
- `useRepoCatalog()` — returns `bySlug`, `presence`, and `refreshPresence()` for catalog-driven key-presence indicators.
- `React.useRef<number | null>(null)` stored as `pageExecIdRef` to track the execution launched from this page instance.
- `React.useEffect` watching `pageExec?.status`: when the tracked execution leaves `running` / `cancelling`, `pageExecIdRef.current` is cleared so the page is ready for a new run.

UI State Model

Local state (`useState`):

State	Type	Purpose
`ephemeralKeys`	`Record<string, string>`	Per-credential-name ephemeral key buffer; only non-empty entries are sent on submit.
`repositories`	`string[]`	Selected repository slugs.
`dateFrom`, `dateTo`	`string`	Optional ISO date strings.
`keywords`	`string[]`	Keyword chips.
`maxResults`	`number`	Per-repository cap (10–500 step 10, default 100).
`aiEnabled`	`boolean`	AI summarization toggle.
`aiOverrideLength` / `aiOverrideTone` / `aiOverrideModel`	`string`	IMPL-AI13 per-execution overrides; empty means "use app default".
`error`	`string`	Inline form error (repo/keyword validation or submit failure).
`saveModalOpen`, `configName`, `saveStatus`	—	State for the Save Configuration modal.
`configRefresh`	`number`	Bump counter that re-triggers `ConfigLoader` after a save.

Refs:

- `pageExecIdRef: React.RefObject<number | null>` — tracks the ID of the execution launched from this page so the result card and the "Run Another" button label reflect the correct run.

Derived values:

- `pageExec` — looked up as `activeExecutions[pageExecIdRef.current]`.
- `running` — true when `pageExec?.status === 'running'`.
- `buildQuery()` — `keywords.join(' ')`.

Key Interactions and Events

- **Form submit (`handleRun`).** Validates `repositories.length ≥ 1` and `keywords.length ≥ 1`; assembles the request body; filters `ephemeralKeys` to non-empty values and sends them as `ephemeral_credentials`; includes `ai_settings` only when at least one of `length` / `tone` / `model` overrides is non-empty; POSTs to `/api/search/sweep`; on success, stores the returned `execution_id` in `pageExecIdRef` and calls `startExecution(id, 'deep_sweep', repositories)` so the shared execution context picks it up.
- **Config load (`applyConfig`).** Populates `repositories`, `keywords`, `maxResults`, and `aiEnabled` from the loaded configuration. `dateFrom` and `dateTo` are intentionally **not** restored.
- **Save Configuration (`handleSaveConfig`).** POSTs to `/api/configurations` with `config_type: "manual_sweep"` and a `parameters` payload containing `repositories`, `date_from`, `date_to`, `keywords`, `max_results`, and `ai_enabled`. On success, closes the modal, clears the name, bumps `configRefresh`, and shows a transient success message that clears after 3 seconds.
- **Ephemeral key entry.** `RepoKeyStatus` invokes `onEphemeralChange(v)`, which updates `ephemeralKeys[credential_name]`. `onPresenceChange` calls `refreshPresence()` from `useRepoCatalog` to re-fetch presence status.
- **Execution status transition.** The `useEffect` on `pageExec?.status` resets `pageExecIdRef` when the tracked execution leaves the running set, enabling a subsequent run from the same page without a reload.

Error and Empty States

- **Empty repositories selection.** Submit sets `error = 'Please select at least one repository.'`.

- **Empty keywords.** Submit sets `error = 'Please enter at least one keyword.'` .
- **Submit exception.** The `catch` block in `handleRun` sets `error` to the caught message (e.g. the 429 from the admission controller) or the generic fallback `'Failed to start sweep.'` .
- **Completed/failed result card.** When `pageExec` exists and its status is neither `running` nor `cancelling` , a `result-card` is rendered. The CSS class is `result-success` when `status === 'completed'` and `result-failure` otherwise; it shows the execution ID, status, result count, new count, elapsed seconds (when available), and a **View Report** link pointing to `#/results?exec=<id>` .
- **No per-repository key status.** The Key Status panel is rendered only when `repositories.length > 0` .

Backend

API Endpoints

Endpoints used by this page:

- `POST /api/search/sweep` — launches a deep-sweep execution.
- `POST /api/configurations` — used by the Save Configuration modal (`config_type: "manual_sweep"`).
- `GET /api/repositories/catalog` and `GET /api/credentials` — consumed through `useRepoCatalog()` / `RepoKeyStatus` for presence state.

Request and Response Patterns

`POST /api/search/sweep` is handled by `search_sweep(body: SweepRequest)` in `resmon_scripts/resmon.py` . `SweepRequest` is the following Pydantic model:

```python
class SweepRequest(BaseModel):
    repositories: list[str]
    query: str
    keywords: Optional[list[str]] = None
    date_from: Optional[str] = None
    date_to: Optional[str] = None
    max_results: int = 100
    ai_enabled: bool = False
    ai_settings: Optional[dict] = None
    ephemeral_credentials: Optional[dict[str, str]] = None
    saved_configuration_id: Optional[int] = None
```

The optional `saved_configuration_id` is set by the page when the user launched the run from a `ConfigLoader`-restored saved config (tracked in `loadedConfigIdRef`). The handler forwards it to `engine.prepare_execution` so the resulting `executions` row links back to the originating saved config; downstream surfaces (Recent Activity, Results & Logs, Calendar popover) render the matching saved-config name without an extra fetch.

The handler:

1. Calls `_reject_if_at_manual_cap()`. If `admission.try_admit(kind="manual")` returns false, FastAPI raises `HTTPException(status_code=429, …, headers={"Retry-After": "5"})` with a detail message referencing **Settings** → **Advanced**.
2. Builds a `SweepEngine` with `config={"ai_enabled": body.ai_enabled, "ai_settings": body.ai_settings}`.
3. Calls `engine.prepare_execution("deep_sweep", body.repositories, query_params)` to insert the execution row and receive `exec_id`.
4. Calls `progress_store.register(exec_id)` and then `_launch_execution(engine, exec_id, conn, ephemeral_credentials=body.ephemeral_credentials)`.
5. Returns `{ "execution_id": exec_id }`.

Response shape: `{ execution_id: number }`. All further state arrives via progress events and the execution record.

`POST /api/configurations` accepts a `ConfigCreate` (`name`, `config_type`, `parameters`) and is used unchanged by the Save Configuration modal.

Persistence Touchpoints

- **`executions` table.** `SweepEngine.prepare_execution` inserts a row with `execution_type = "deep_sweep"`, a JSON-serialised `parameters` dict

(which includes `repositories`), `start_time`, and — when the request carried a `saved_configuration_id` — the foreign key linking the execution to its originating saved config. The row is later updated by `update_execution_status` with `end_time`, `result_count`, `new_result_count`, `log_path`, `result_path`, and (on full failure) an `error_message`.

- **`documents` table.** For each raw result, `normalize_result` + `validate_result` + `get_document_by_source` / `link_execution_document` persists document rows and creates execution ↔ document associations. `first_seen_at` is compared against the execution's `start_time` to compute the `new_count`.
- **`configurations` table.** `POST /api/configurations` writes a row with `config_type = "manual_sweep"` and the JSON `parameters` payload.
- **`app_settings`.** Read in `_apply_ai_settings_to_engine` to obtain the persisted `ai_*` keys listed in `_AI_SETTING_KEYS`.
- **Progress events.** Emitted in memory during the run via `progress_store.emit`; `save_progress_events(conn, exec_id, events)` persists the event stream into the database at the end of `_launch_execution`.

Execution Side Effects

`_launch_execution` starts a daemon thread that runs the full pipeline; the HTTP request returns immediately. The thread:

1. **Admission bookkeeping** — `admission.note_admitted(exec_id)` on entry, `admission.note_finished(exec_id)` in the `finally` block.
2. **Ephemeral credential registration** — `push_ephemeral(exec_id, ephemeral_credentials or {})` so that `get_credential_for(exec_id, name)` used inside API clients sees the per-execution keys; `pop_ephemeral(exec_id)` runs in `finally`.
3. **AI settings resolution (`_apply_ai_settings_to_engine`).** Loads persisted `ai_*` settings and merges them with the per-execution override (`engine.config["ai_settings"]`), after translating IMPL-AI13 short keys through `_AI_OVERRIDE_KEY_MAP` (`length → ai_summary_length`, `tone → ai_tone`, `model → ai_model`). Builds `engine.config["ai_prompt_params"]` (length, tone, extraction_goals, `_show_audit_prefix`, `_audit_provider`, `_audit_model`). Calls `build_llm_client_from_settings`. If `ai_enabled` is true but the factory returns `None` or raises `ValueError`, a single `log_entry`

warning progress event is emitted (`"AI skipped: …"`) and the execution continues with `engine.llm_client = None` .

4. **Pipeline execution (`SweepEngine.run_prepared(exec_id)`)** — see the stage list below.
5. **Progress finalisation.** `progress_store.mark_complete(exec_id)` → persist events via `save_progress_events` → `progress_store.cleanup(exec_id)` .
6. **Routine completion email hook (IMPL-R7).** Not fired for manual sweeps because the branch is gated on `execution_type == "automated_sweep"` and a non-null `routine_id` .

Sweep pipeline stages (inside `SweepEngine._run` , driven by `run_prepared`):

1. **`execution_start` event** with the repositories list and `total_repos` .
2. **`stage = "querying"`** — iterate `repositories` :
   - Emit `repo_start` ; consult `_REQUIRED_CREDENTIALS` and, if the repository needs a key that is not available, emit `repo_skipped_missing_key` and append to `missing_key_repos` (the query still runs, but the upstream client typically returns zero results).
   - Call `get_client(repo_name)` ; set `client._exec_id = exec_id` so the client can look up ephemeral credentials via `get_credential_for` .
   - Run `_search_with_heartbeat(...)` , which performs the HTTP query subject to each client's built-in rate limiting / retry logic and emits `heartbeat` events while the call is in flight.
   - Emit `query_progress` (`search_start` , `search_done`) and `repo_done` with the per-repo result count. Errors are caught, logged, appended to `repo_errors` , and emitted as `repo_error` events; the sweep continues with the next repository.
   - Cancellation is checked at every iteration via `progress_store.should_cancel(exec_id)` .
3. **`stage = "dedup"`** — `deduplicate_batch(self.db, all_results)` runs cross-source deduplication and returns counts (`total` , `new` , `duplicates` , `invalid`), emitted as `dedup_stats` .
4. **`stage = "linking"`** — for each raw result, normalise/validate, look up or create the `documents` row, and link it to the execution via `link_execution_document` . Progress is emitted as `link_progress` at ~5% granularity.
5. **`stage = "reporting"`** — `_build_report_docs(all_results)` assembles the per-document dicts used for the report.

6. **Optional `stage = "summarizing"` (only when `self.llm_client` is set, `ai_enabled` is true, and there are documents to summarize).** Iterates documents one at a time so cancellation and per-document progress can be honoured; `SummarizationPipeline.summarize_document(text)` is invoked per entry with the prompt params built earlier. Failures on a single document are logged and skipped (the batch continues); an `ai_summary` key is added to each successful document dict. Emits `ai_start`, `ai_progress` per document, and `ai_done`.
7. **Report generation.** `generate_report(report_docs, report_metadata)` produces Markdown including the query, keywords, repositories, `missing_key_repos`, date range, totals, and (when set) the `provider/model` audit label. The report is saved to `REPORTS_DIR/markdowns/report_deep_sweep_<exec_id>_<ts>.md`; a `report_saved` event is emitted.
8. **`stage = "finalizing"`** — `update_execution_status` writes final counts, `log_path`, `result_path`, and status. Status is `failed` only when *every* requested repository errored (`all_repos_errored`); otherwise it is `completed`, even if a subset failed (their errors remain in the task log and as `repo_error` events).
9. **Auto-backup.** `_maybe_auto_backup(...)` spawns a daemon thread to optionally push the report and log to cloud storage; failures are logged but never fail the execution.
10. **Final events.** `complete` event (with `status`, `result_count`, `new_count`, `elapsed`, and — on full failure — the `error` summary), followed by `progress_store.mark_complete(exec_id)`.

Cancellation. `_handle_cancellation` is invoked if the cancel flag is set at any inter-stage checkpoint or between documents in the AI step; it writes a partial report from `all_results` collected up to that point, updates the execution row to `cancelled`, and emits the terminal events.

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